

A non-equilibrium statistical model-based constitutive equation for predicting creep lifetime of alloys

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2025.08.10. Kanazawa, Japan

12th JPN-CHN Bilateral Symposium



The Challenge for Creep Lifetime Prediction

TMSR Nuclear Reactor

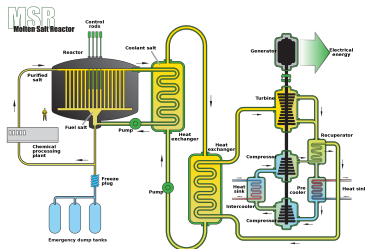
- Nuclear Power
→ 30MWt
- Temperature > 650°C
- Neutron Irradiation
> 10dpa
- Corrosion (F, Li, Be)

Structural Materials

- GH3535 Nickel-based Super Alloy
- Service Lifetime $\geq$ 10yrs
- Creep Properties → Strength Calibre

Creep Experiments

- High Time Consumption
- Experimental Expense (Irradiation Sample)
- Creep in Hot Cell ($\leq$ 1000hrs)



The Question:

How to predict the creep lifetime ?

Short-Time Data → Long-Time Creep Life

J.Nucl.Mater.**527**, 151837 (2019), Eng.Fail.Anal.**134**, 105968(2022)

The Equations for Predicting Creep Lifetime

- Larson-Miller equation:

$$P_{LM} = f(\sigma) = T(C_{LM} + \log t_f)$$

$P_{LM} \Rightarrow$ fitting parameters

Larson-Miller constant: $C_{LM} \sim 20$

- Manson-Haferd equation:

$$P_{MH} = T(\log t_f - \log t)/(T - T_a)$$

P_{MH} , $T_a \rightarrow$ fitting parameters

- Orr-Sherby-Dorn:

$$P_{OSD} = (\log t_f - C_{OSD})/T$$

P_{OSD} , $C_{OSD} \rightarrow$ fitting parameters

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Monkman-Grant relationship:

$$\dot{\epsilon}_{ss} \times t_f = K$$

$$\dot{\epsilon}_{ss}^m \times t_f = K'$$

J.Nucl.Mater.**581**.154441(2023)

[Fe-steel]

Mater.Sci.Eng.A **845**.143027(2022)

[25Cr/35Ni]

Metall.Mater.Trans.**30A**.829(1999)

[DS pure-Al, Al-alloy]

J.Mater.Eng.Perform.**1**.123(1992), [Cr-Mo]

Metal Sci.**10**.382(1976), [Ni-4.8%Al]

Key Questions:

Creep Constitutive Equation $\rightarrow \dot{\epsilon}_{ss} \sim T, \sigma?$

Creep Mechanisms and Constitutive Equations

Microscopic Mechanisms:

- Dislocations
 - Gliding
 - Climbing(Harper-Dorn)
- Nabarro-Herring Creep (Diffusion)
- Coble Creep (Grain Boundary Diffusion)
- Solute Drage Creep (Dislocations + Atomic Diffusion)

Eng.Failure.Analy.**134**.105968(2022),
 Prog.Mater.Sci.**45**.1(2000),
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 Mater.Sci.Eng.A **776**.139036(2020),
 J.Appl.Phys.**123**.145108(2018))

Creep Constitutive Equations:

- ① **Power-Law Equations**(Widely Used):

$$\dot{\varepsilon}_{ss} = A\sigma^n \exp[-Q_c/k_B T]$$
 - $n \simeq 5$, $Q_c \sim Q_{\text{diffusion}}$ (High-T/ σ Breakdown)
- ② Dorn Equation: $\dot{\varepsilon}_{ss} = \frac{AD\mu b}{k_B T} \left(\frac{\sigma}{G}\right)^n$
- ③ Rosler-Arzt Equation (Disperse Strengthen):

$$\dot{\varepsilon}_{ss} = \frac{6D_v\lambda\rho}{Mb} \exp\left[-\frac{\mu b^2 r}{k_B T} (1-k)^{3/2} \left(1 - \frac{\sigma}{\sigma_D}\right)^{3/2}\right]$$
 - $k, \rho \rightarrow$ (Dislocation Density)
- ④ Microstructure-Based Equation:

$$\frac{\dot{\varepsilon}_{ss} k_B T}{D_L G b} = \frac{\pi\Omega(k_B T)^2}{b(\gamma M)^2} \left(\frac{\sigma}{G}\right)^3 (A\sigma^2 + B\sigma + C) \left[\frac{D_{sol}}{2\pi c_0 \beta^2 \ln(r_2/r_1) D_L + (b k_B T)^2 \ln(c^*/c_0^*)} \right]$$
- ⑤ SSTC Equation:

$$\varepsilon = \varepsilon_0 + A_1(1 - e^{-A_2 t}) + A_3(1 - e^{-A_4 t^3})$$

Creep Mechanisms and Constitutive Equations

Microscopic Mechanisms:

- Dislocations
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- Solute Drag Creep (Dislocations + Atomic Diffusion)

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Phys.Rev.Lett.**132**.256201(2024),Metals **9**.342(2019)
Mater.Sci.Eng.A **776**.139036(2020),
J.Appl.Phys.**123**.145108(2018))

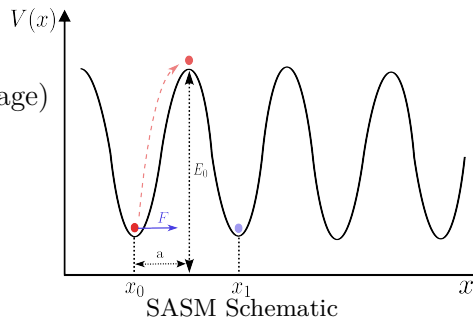
A Qualified Equation with Fitting Parameters:

- A. **Accuracy**
- B. **Universality**
- C. **Simple Mathematical Form**
- D. **Unambiguous Physics**
- E. **Practical Feasibility**

Single-Atom Statistical Model(SASM)

- Non-Equilibrium Statistical Mechanics
 - Kinetic Energy $\longleftrightarrow$ Maxwell Velocity Distribution
 - $f(\varepsilon) = \varepsilon^{\frac{1}{2}} \exp[-\frac{\varepsilon}{k_B T}]$
- Jumping Rate over Effective PES (Ensemble Average)

$$\begin{aligned} \Gamma &= \mathcal{P}|_{\varepsilon > E_0} \times \frac{1}{\langle \delta t \rangle |_{\{\varepsilon\}}} \\ &= \frac{[\int_{E_0 - F \cdot a}^{\infty} \varepsilon^{\frac{1}{2}} \exp[-\frac{\varepsilon}{k_B T}] d\varepsilon]^2}{\int_0^{\infty} \varepsilon^{\frac{1}{2}} \exp[-\frac{\varepsilon}{k_B T}] d\varepsilon} \times \\ &\quad \frac{1}{\int_{E_0 - F \cdot a}^{\infty} \int_{x_0}^{x_1} \frac{\sqrt{m}}{\sqrt{2(\varepsilon - V(x))}} dx \varepsilon^{\frac{1}{2}} \exp[-\frac{\varepsilon}{k_B T}] d\varepsilon} \end{aligned}$$



- SASM Creep Equation: $\dot{\varepsilon}_{ss} = K \cdot a \cdot \Gamma$
 - Fitting Parameters : E_0, a, K (*Independent* on T/σ)
 - $E_0 \rightarrow$ PES Height, $a \rightarrow$ PES Width,
 - $K \rightarrow$ Number of Involved Units

Europhys.Lett.**94**.40002(2011)
ibid,**95**.47012(2011)
 Phys.Rev.E **87**.062311(2013)
 Chin.Phys.B **24**.093401(2015)
 Appl.Surf.Sci.**470**.448(2019)

Modified Single-Atom Statistical Model

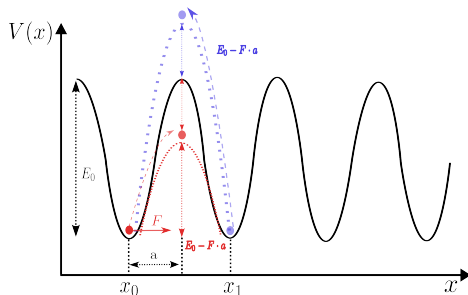
● Bidirectional Jumping Rate (Along/Anti Stress Direction)

$$\begin{aligned}\Gamma_{\text{forward}} &= \mathcal{P}|_{\varepsilon > (E_0 - F \cdot a)} \times \frac{1}{\langle \delta t \rangle |_{\{\varepsilon\}}} \\ \Gamma_{\text{backward}} &= \mathcal{P}|_{\varepsilon > (E_0 + F \cdot a)} \times \frac{1}{\langle \delta t \rangle |_{\{\varepsilon\}}} \\ &= \frac{[\int_{E_0 + F \cdot a}^{\infty} \varepsilon^{\frac{1}{2}} \exp[-\frac{\varepsilon}{k_B T}] d\varepsilon]^2}{\int_0^{\infty} \varepsilon^{\frac{1}{2}} \exp[-\frac{\varepsilon}{k_B T}] d\varepsilon} \times \frac{1}{\int_{E_0 + F \cdot a}^{\infty} \int_{x_0}^{x_1} \frac{\sqrt{m}}{\sqrt{2(\varepsilon - V(x) - F \cdot a)}} dx \varepsilon^{\frac{1}{2}} \exp[\frac{-\varepsilon}{k_B T}] d\varepsilon}\end{aligned}$$

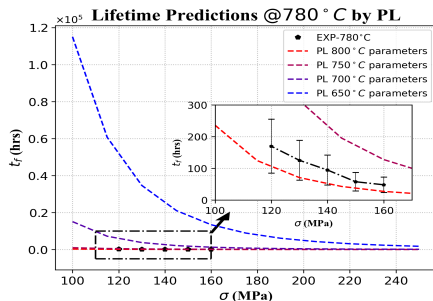
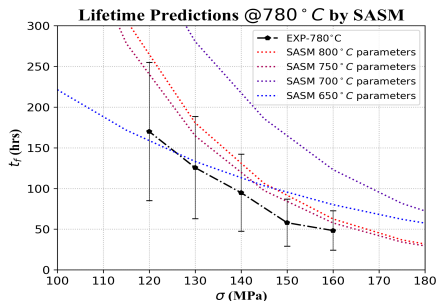
● Modified SASM Creep Equation:

$$\dot{\varepsilon}_{ss} = K \cdot a \cdot [\Gamma_{\text{forward}} - \Gamma_{\text{backward}}]$$

- Fitting Parameters : E_0, a, K
(Independent on T/σ)
- $E_0 \rightarrow$ PES Height, $a \rightarrow$ PES Width,
 $K \rightarrow$ Number of Involved Units

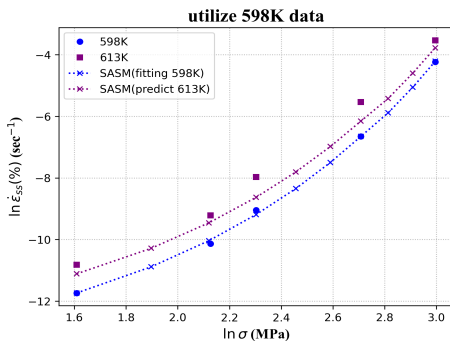
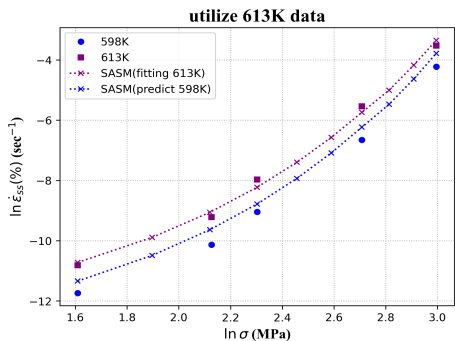


GH3535-Alloy Creep by Modified-SASM Equation



- Short-Time Data → Long-Time Creep Time
- Parameters Determined by Different Data Sets
- Prediction Precision (about an order better than Power-Law)

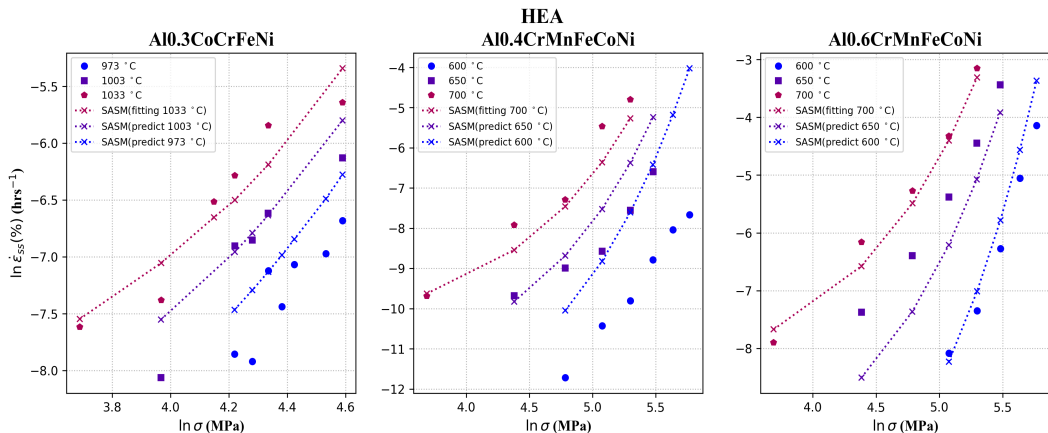
Pure Al-Metal Creep by Modified-SASM Equation



Fitting Temperature	E_0	a	K
598K	1.463	5.224×10^2	1.283×10^2
613K	1.458	5.247×10^2	1.696×10^2

Phil.Mag. 26, 1419(1972)

High-Entropy-Alloy Creep by Modified-SASM Equation



Acta Mater. **244**.118600(2023), Mater.Sci.Tech. **35**.1283(2019)

Conclusion and Outlook

Conclusional Remarks

- ① Non-Equilibrium Statistical Mechanics→Creep Constitutive Equation
- ② Understanding Creep Dynamics from The Perspective of an Effective PES
- ③ Long-Time Creep Derived from Short-Time Data
- ④ SASM Creep Model: Precision, Universality、Feasibility

Outlook

- ① Full-Creep Process (First, Stationary and Tertiary)
- ② Irradiation Effects

Thanks for your attention!
Comments are MOST welcome!